

MSE 609
Project Report
Health Insurance Charge Analysis

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Introduction

With the commodification of health care services, access to health insurance becomes essential for citizens to be protected from unexpected financial loss due to illnesses and injuries, especially for low-income populations. While the Canadian government provides health plans to cover citizens’ basic medical services, many people still need private insurance to cover additional healthcare needs. In contrast, countries like the United States, which lacks universal healthcare, rely entirely on private insurance markets. Insurance companies calculate rates for individuals by assessing risks based on factors such as age, sex, and lifestyle. Among these, the habit of smoking is considered an important factor as it is associated with many negative health consequences – people who smoke are about 25 times more likely than non-smokers to develop lung cancer(National Center for Chronic Disease Prevention and Health Promotion (US) Office on Smoking and Health, 2014), and smoking increases the risk of coronary heart disease and stroke by 2 to 4 times (CDC, 2024). Research also shows that smoking leads to excess medical care where the medical costs for male smokers are 11% more than non-smokers (Izumi et al., 2001). Insurance companies rely on these risk factors to assess the potential medical expenses they need to cover and determine the insurance prices for beneficiaries.

Given these considerations, we want to investigate how smoking impacts the medical charges covered by insurance companies and explore other factors that might contribute to the variation of the amount of charges paid by insurance. This will provide valuable insights into how lifestyle choices, particularly smoking, influence the cost of health insurance and the extent of coverage. This information can help individuals make informed decisions about their health behaviors and financial planning. This study can also inform policymakers and insurance providers about the equity and efficiency of insurance pricing models.

Data Description

Data Source:

The data for our research is sourced from the *Medical Cost Personal Datasets* on Kaggle, a platform that is widely used for exploring machine learning models (*Medical Cost Personal Datasets*, n.d.). This dataset originating from the book ‘Machine Learning with R’ was collected by Miri Choi in 2017. While it can only be accessed by purchasing the book, Miri Choi made it open-access for data enthusiasts to use. Kaggle evaluates the usability of each dataset with strict criteria of completeness, credibility, and compatibility. Only individuals with proper licenses have permissions to publish datasets. For our dataset, Miri Choi has a database contents license and has marked this dataset for public use. The dataset used in our research achieved a high score of usability, which provides detailed demographic, habit, health, and family information of beneficiaries. It includes variables such as the number of children the beneficiaries have, smoking status, sex, gender, age, BMI, and medical charges.

Key Variables:

#	Predictor	Description	Value	Data Type
1	Age	Age of primary beneficiary	18 - 64	Integer
2	Sex	Gender of primary beneficiary	Male/Female	Categorical
3	BMI	Body Mass Index	15.96 – 53.13	Float

		Estimating body fat by dividing a person's weight in kilograms by the square of the height in meters (kg/m ²)		
4	Children	Number of dependents covered by insurance	0 - 5	Integer
5	Smoker	Whether primary beneficiary has a habit of smoking	Yes/No	Binary
6	Region	Area where beneficiary live in	Southwest/Southeast/ NorthWest/NorthEast	Categorical
7	Charges	Individual's medical expenses billed by health insurance	1121.874-63770.43	Float

Table 1.

Descriptive Data Analysis

1. Univariate analysis:

- Age: the mean age of the sample beneficiaries is 39.21, with the youngest being 18 and the oldest being 64.
- Sex: the proportions of females and males are about the same, where the female is 49.5% and males is 50.5%.
- Children: most beneficiaries have no or only 1 child, and the maximum number of children in our sample is 5.
- Smoker: 70.5% of the beneficiaries identify as smokers and 20.5% as non-smokers.
- Region: proportions of beneficiaries living in four regions (north/southwest, north/southeast) are about the same.
- BMI: values of BMI have a fairly normal distribution and center around the mean of 30.66. Most beneficiaries have a BMI between 26 and 34.
- Charges: the medical charge billed by insurance for most beneficiaries ranges from \$1,122 to \$14,000. A few outliers exceed the charge of 50,000 with a maximum of 64770.

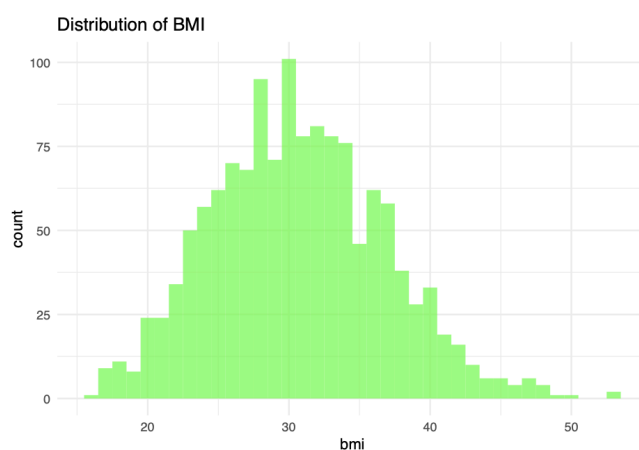


Figure 1.

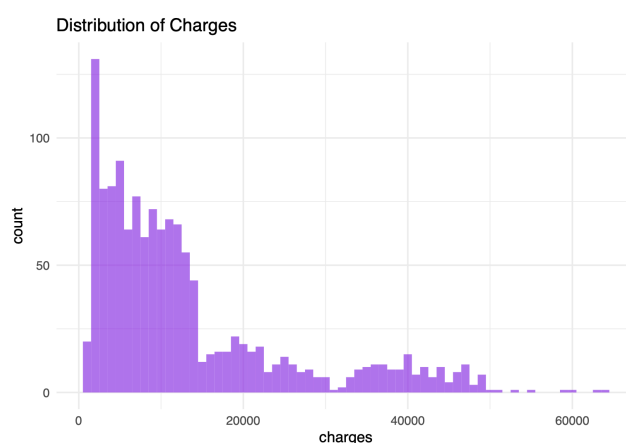


Figure 2.

2. Bivariate analysis:

a) Correlation between charges and predicting variables:

As shown by the scatter plot of Charges vs. Age, there is a positive trend between a beneficiary's age and the medical charges. The charges increase as age increases. Notably, the data points are stratified, which will be further examined in section b). The plot for Charges vs. BMI also shows a positive relationship where the charges increase as BMI increases. Some data points cluster at the bottom below the regression line, which will also be further explained in b).

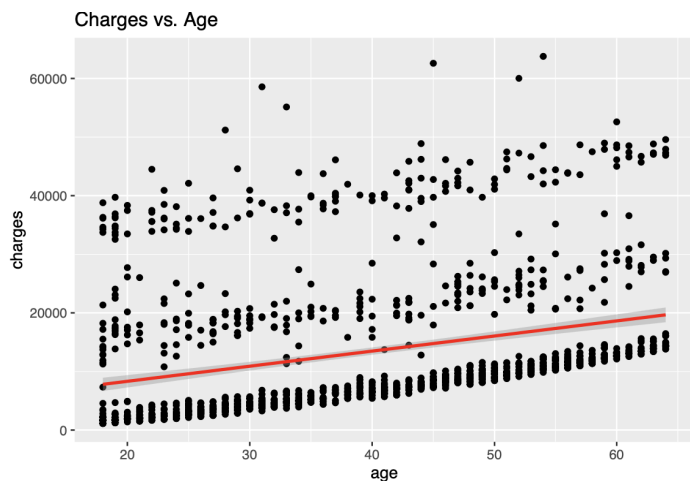


Figure 3.

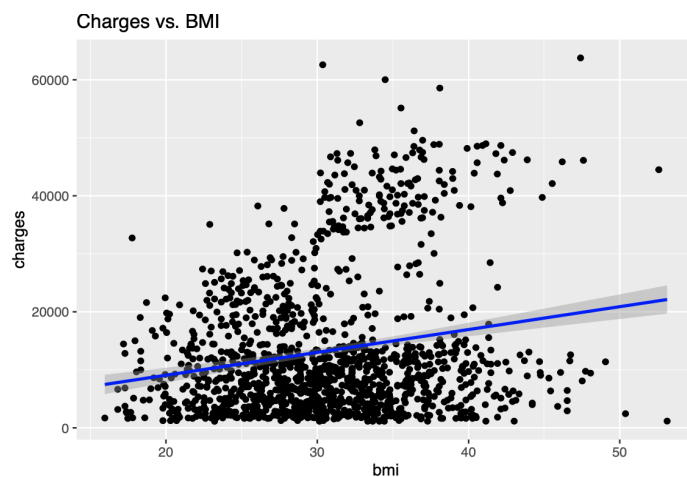


Figure 4.

The plot that compares charges for smokers and non-smokers indicates that charges are significantly higher for smokers as the whisker and median for smokers are significantly higher than that of non-smokers. The plot for Charges by Sex shows that the median charges for females and males are about the same, while the whisker for males is considerably larger than the whisker of females, which means males have a higher 3rd quantile charge.

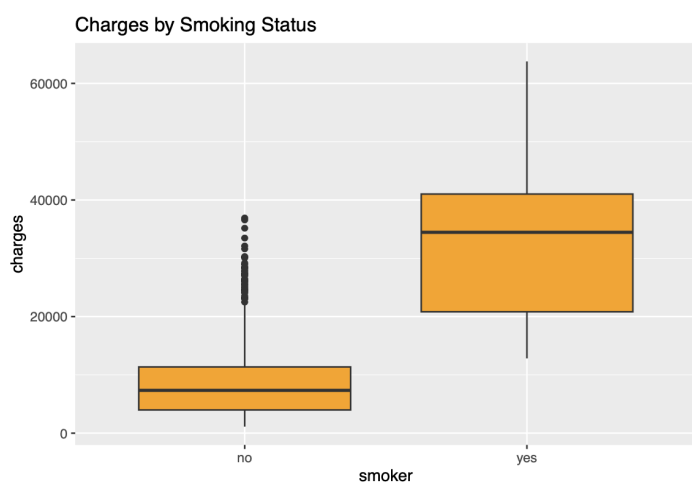


Figure 5.

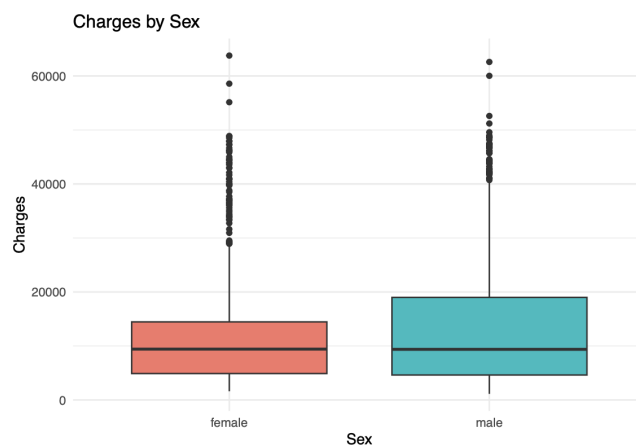


Figure 6.

The plot for Charges by Region shows that the median charges for all four regions are about the same, with southwest and northeast having higher 3rd-quantile charges. The plot for Chagres by Children also indicates no obvious trend as the medians across different numbers of children are about the same.

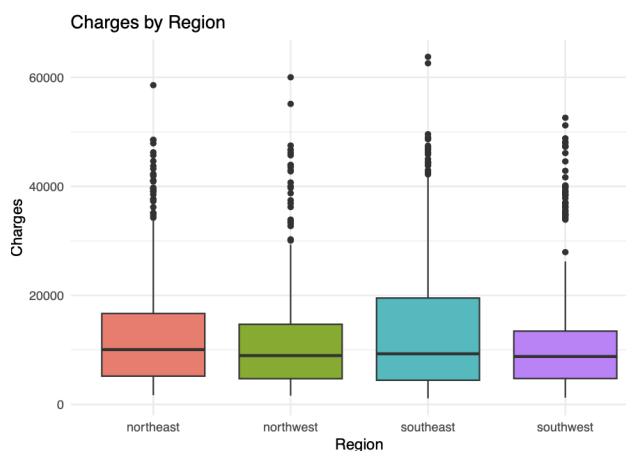


Figure 7.

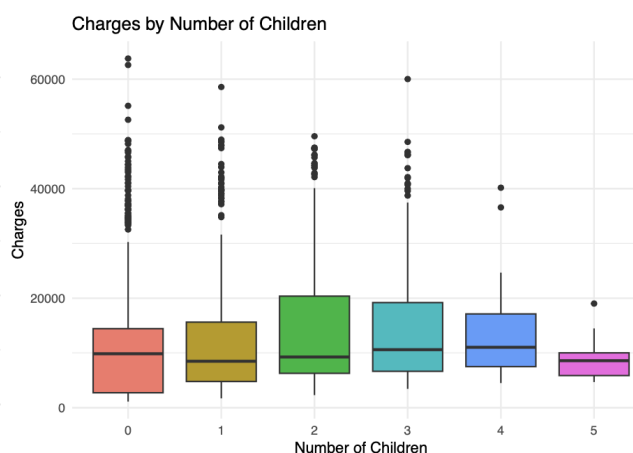


Figure 8.

b) Correlations among predictors:

To more effectively analyze the relationship between predictors and medical charges billed by insurance, we first examined the independence among numeric predictors by testing their inter-variable correlations. The correlation matrix

plot shows that the correlations among age, BMI, and number of children are weak, which suggests that these variables can be considered independent.

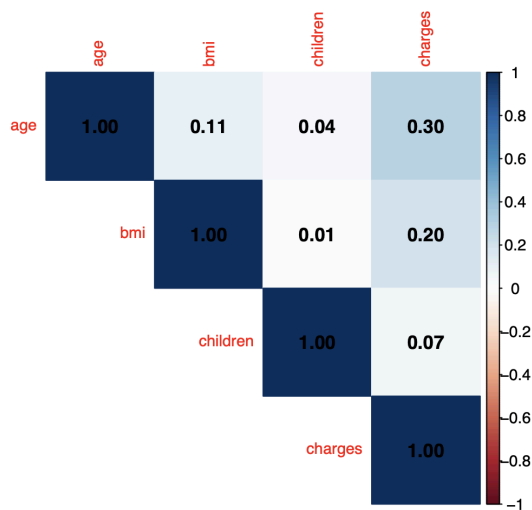


Figure 9.

To further investigate the stratification and clustering discovered in Figure 3 and 4 in section a), we color-coded the data points by smoking status in the plots for Charges vs. Age and Charges vs. BMI. The color scheme shows that the clusters correspond to smoker status. In Charge by Age and Smoking Status, the lowest group of charges (the bottom strata) are primarily non-smokers. Similarly, in Charges by BMI and Smoking Status, the cluster with lower charges are all non-smokers. The plots also indicate that the effects of age and BMI on medical charges do not depend on smoking status as there are no obvious patterns of correlation between smoking and BMI/age.

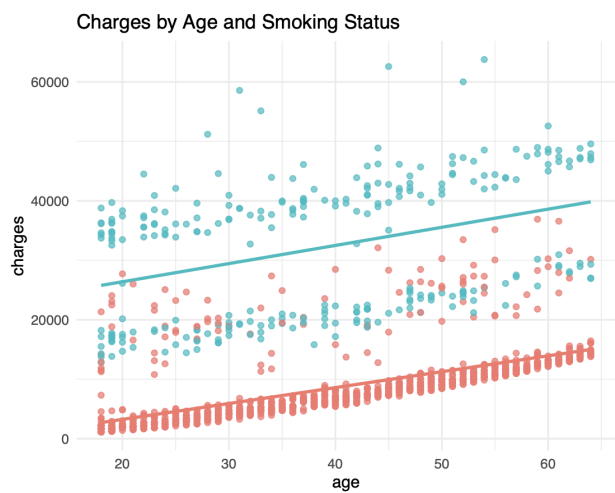


Figure 10.

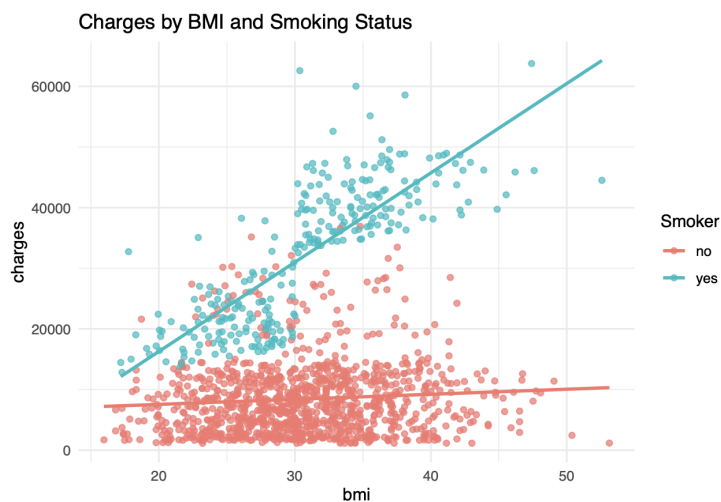


Figure 11.

The stacked bar plot explores how changes vary for smokers and non-smokers among females and males. It shows that there are more smokers among males in general and the medical charges are higher for males. This implies that the relationship between sex and charges may be affected by one's smoking status.

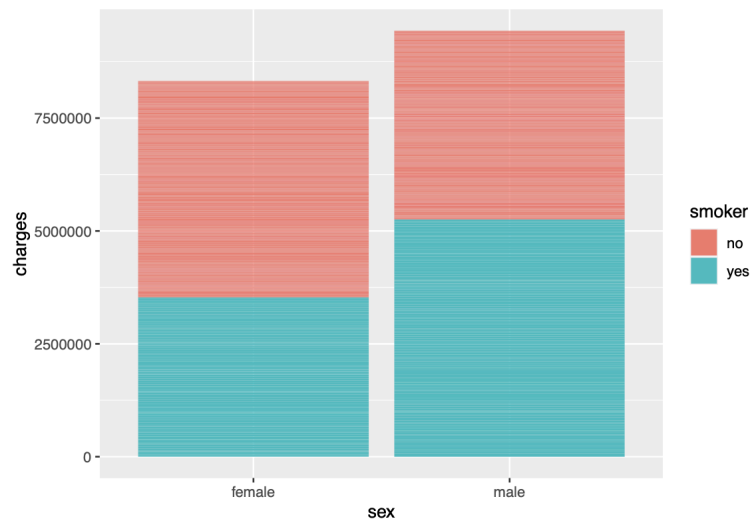


Figure 12.

Question of Interest

Primary Research Question:

- How does being a smoker affect a person's medical expenses?

Exploratory Question:

- What are the top three leading factors that influence medical costs billed by insurance?

Data Relevance:

To address our research question, we need to compare the medical costs of people who smoke with those who do not. The variable of smokers indicates the smoking status of individuals, and the variable charges represent their medical charges billed by health insurance, allowing us to clearly identify the relationship between those two. Based on data from other predictors such as BMI, age, number of children, and sex, medical institutions are able to gauge the risks of potential diseases for each person and charge differently. This helps us determine the primary contributing factors to these costs. Insurance claims vary across regions due to differences in taxes and standards of medical systems. Analyzing this parameter can provide an insight into how regional factors impact medical expenses.

Initial Model and Estimation

1. Data processing

For the prediction, we split our dataset into testing and training with a ratio of 2:8. In order to have the regression algorithms correctly process the data, we made the following conversions on variables:

- Binary categorical variables like sex and smoking are converted to binary numeric variables, 0 and 1.
- Continuous variables are standardized to have a mean of zero and a standard deviation of 1. This standardization is crucial for Ridge and Lasso regression as they are sensitive to the scale of input features.
- Categorical data with k-level like Region is converted to dummy variables.

2. Linear Regression

a. Model selection

For the analysis, we chose Linear regression as the primary analytical tool to investigate the factors influencing charges. It is suitable for this dataset and initial statistical model for several reasons:

- Linear regression allows for the prediction of a continuous outcome for multiple predictor variables. This model helps quantify the relationship between each predictor and the outcome, which is crucial for

understanding which factors contribute to the individual medical costs billed by health insurance the most.

- The model provides the coefficient for each predictor, representing the expected change in the dependent variable for a one-unit change in the independent variable, holding the other variable constant. This allows valuable insights for policy formulation and strategic planning.

b. Estimation Strategy

The estimation of the linear regression model was performed using the least squares method. This method minimizes the sum of the squared differences between the observed value (actual charges) and the predicted value by the model. Therefore, the best possible fit of the data under the linear assumption is ensured. Given that our dataset does not exhibit exponential relationships or significant outliers, a transformation such as log was unnecessary for our analysis. Therefore, the linear model is applied directly.

Our regression is as follows:

$$Charges = B_0 + B_1age + B_2sex + B_3BMI + B_4children + B_5Smoker + B_6Region + \varepsilon$$

The regression aims to quantify how demographic and health-related factors like age, sex, BMI, the number of children, smoking status, and geographical region affect individual medical costs billed by health insurance.

3. Ridge and Lasso Regression

a. Model Selection

We proceed with Ridge and Lasso regression to enhance the model. We chose them because both of them allow us to enhance the model by addressing potential multicollinearity and overfitting. Ridge regression effectively reduces variance without substantially increasing bias, which improves model stability. It is ideal when all predictors have a small to moderate influence on the outcome. It regularizes the coefficients by shrinking them towards zero but not exactly zero, thereby maintaining all the features but reducing their impacts. On the other hand, Lasso offers the advantages of feature selection by setting the coefficient of less relevant predictors to zero. It introduces more bias than ridge but with a trade-off of potentially lowering variance. By applying both regressions to our model, we will be able to leverage their respective strengths to achieve optimal performance of our model by balancing between complexity and interpretability.

b. Estimation Strategy

- Ridge implementation involves the use of penalty parameter lambda, which helps us control the extent of coefficient shrinkage. The optimal lambda is tuned using the cross-validation techniques to determine the optimal penalty strength. This process ensures that the coefficients are shrunk appropriately.
- Similar to the Ridge, Lasso regression uses lambda parameters to regulate the impact of the penalty. However, it allows us to do feature selection by forcing some unimportant coefficients to zero, which leads to a simpler model. The lambda is tuned using cross-validation techniques which ensure the model avoids overfitting while maintaining good predictive performance.

Interpretation of Model Results

The interpretation of the models' results will help us answer the research and exploratory question. It allows us to understand model performance and significance of each predictor. As the data has been scale, comparing the coefficients allows us to understand the relative importance of each predictor.

1. Result summary of linear regression

Figure 13 is the summary of the linear regression. It shows:

- Variance explained: The adjusted R-squared indicates the variance explained. In our linear model, we have 0.7576. This means approximately 76% of the variance in medical charge is explained by the model.
- F-statistic: The p-value of the F-statistic is highly significant ($< 2.2e-16$). This confirms that the model is significant.
- Sex and Region are not statistically significant as indicated by the large p-value. They may be excluded from the model.
- Smoker, BMI, and Age seem to be more important predictors. Out of the three, the coefficient of smoker is significantly higher than the other predictor, indicating that smoking status has the most substantial impact on dependent variables.

```
Call:
lm(formula = trainY ~ ., data = as.data.frame(trainX))

Residuals:
    Min       1Q   Median       3Q      Max
-0.97026 -0.22641 -0.07582  0.12009  2.44490

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.43985    0.03518  -12.504 < 2e-16 ***
age             0.28668    0.01536   18.667 < 2e-16 ***
sex            -0.00965    0.03046   -0.317  0.75143
bmi             0.16496    0.01597   10.326 < 2e-16 ***
children       0.04356    0.01536    2.836  0.00465 **
smoker         2.00194    0.03760   53.245 < 2e-16 ***
region_northeast 0.08612    0.04428    1.945  0.05206 .
region_northwest 0.04365    0.04419    0.988  0.32344
region_southeast 0.00510    0.04276    0.119  0.90509
region_southwest NA         NA         NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4963 on 1063 degrees of freedom
Multiple R-squared:  0.7594,    Adjusted R-squared:  0.7576
F-statistic: 419.4 on 8 and 1063 DF,  p-value: < 2.2e-16
```

Figure 13. Summary of the Regression Model

2. Result Summary of Ridge Regression

Figure 14 is the estimation summary of Ridge regression, it shows:

- Variance explained: The adjusted R-squared of our Ridge model is 0.712. This means that approximately 71.2% of the variance in individual medical costs billed by health insurance is explained by the model. It is slightly lower than our linear model but still indicates a strong ability to capture the variability in the data.
- Using cross-validation, our optimal lambda is 0.08054.
- Sex and Region (northwest) has been shrunken towards zero. This indicates that these variables have little influence on the insurance charge.
- Age, BMI, and Smoker seem to be more important predictors. Smoker has the highest coefficient of 1.8512. This indicates that smoking is the most influential predictor in the model.


```

[1] "Ridge Coefficients:"
10 x 1 sparse Matrix of class "dgCMatrix"
               s1
(Intercept)    -0.380909219
age             0.265818518
sex            -0.000286307
bmi            0.153558294
children       0.041336555
smoker         1.851236169
region_northeast 0.046776598
region_northwest 0.006294199
region_southeast -0.016383825
region_southwest -0.033489280
[1] "Ridge Lambda (Best Penalty): 0.08054"
[1] "Ridge R-squared: 0.712"

```

Figure 14. Result of Ridge Regression

3. Result Summary of Lasso Regression

Figure 15 is the estimation summary of Lasso regression, it shows:

- Variance explained: The adjusted R-squared of our Lasso model is 0.71. This means that approximately 71% of the variance in individual medical costs billed by health insurance is explained by the model. It is slightly lower than our linear and Ridge model but still indicates a strong ability to capture the variability in the data.
- Using cross-validation, our optimal lambda for lasso regression is 0.00582.
- Sex and Region (Southeast) have been completely removed from the model as their coefficient is forced to zero. This indicates that these predictors do not contribute significantly to our model.
- Age, BMI, and Smoker seem to be more important predictors. Smoker has the highest coefficient of 1.9856. This indicates that smoking is the most influential predictor in the model.

```

[1] "Lasso Coefficients:"
10 x 1 sparse Matrix of class "dgCMatrix"
               s1
(Intercept)    -0.423787812
age            0.282128970
sex            .
bmi            0.156692841
children       0.038106133
smoker         1.985617668
region_northeast 0.055763527
region_northwest 0.013723239
region_southeast .
region_southwest -0.007091423
[1] "Lasso Lambda (Best Penalty): 0.00582"
[1] "Lasso R-squared: 0.71"

```

Figure 15. Result for Lasso Regression

4. Model Comparison

In our analysis, we chose three different regressions - linear, Lidge, and Lasso. Figure 16 is the RMSE of each model. RMSE is calculated by averaging the root mean squared errors of each fold in cross-validation. These errors provide us with a quantitative measure of each model's accuracy.

Model <chr>	RMSE <dbl>
Linear Regression	0.5193958
Lasso Regression	0.5203408
Ridge Regression	0.5186231

Figure 16. RMSE for Model

We anticipated that Ridge and Lasso regression would outperform linear regression as they minimize the impact of less critical predictors and address multicollinearity. However, the result indicated that all models performed similarly in terms of RMSE. This suggests that for our specific dataset, the variables included may not show strong multicollinearity or they are sufficiently informative that the regularization applied in Ridge and Lasso does not lead to better performance significantly.

Ridge regression shows lower RMSE compared to other models. This suggests that its regularization approach might be slightly more effective for our dataset. However, the difference is subtle. This means that the choice between these models might involve the consideration of other factors like interpretability or the importance of feature selection rather than just predictive accuracy.

5. Coefficient Interpretation

For interpretation, we choose to interpret the linear regression. Figure 13 provides insights into the relationship between the predictors and charge. In the analysis, all predictors were standardized. This means each coefficient in the regression output represents that for every one standard deviation change in the predictor, there is an associated expected change in the dependent variable, also measured in standard deviation. This standardization makes it easier to compare the influence of each predictor directly.

- Intercept (B_0): The intercept is -0.43985. This value represents the mean value of the individual medical costs billed by health insurance when all other predictor variables are at their mean.
- Age (B_1): The coefficient for age is 0.28668. This represents that the medical charge will increase by approximately 0.28668 for each additional age, holding other factors constant. The coefficient is highly significant as indicated by the small p-value ($< 2e-16$). This suggests that older individuals are likely to have higher medical costs.
- Sex (B_2): The coefficient for sex is -0.00965. This represents that for males, individual medical costs billed by health insurance will decrease by approximately 0.00965 compared to females, holding all other factors constant. The coefficient is not statistically significant as indicated by the p-value (0.75143). This suggests that sex may not have a meaningful impact on individual medical costs billed by health insurance in the model.
- BMI (B_3): The coefficient for BMI is 0.16496. This represents that the medical charge will increase by approximately 0.16496 for each unit increase in BMI. The coefficient is highly significant as indicated by the small p-value ($< 2e-16$). This suggests that individuals with higher BMI are associated with higher individual medical costs billed by health insurance.
- Children (B_4): The coefficient for children is 0.04356. This represents that the medical charge will increase by approximately 0.04356 with each additional child, holding other factors constant. The coefficient is significant as

indicated by p-value (0.00465). This suggests that having more children leads to higher individual medical costs billed by health insurance.

- Smoking (B_5): The coefficient for smoking is 2.00194. This represents that the medical charge for smokers is 2.00194 compared to non-smokers, holding other variables constant. The coefficient is highly significant as indicated by a small p-value ($< 2e-16$). This suggests that there is a strong effect of smoking on an increase in individual medical costs billed by health insurance.
- Region (B_6)
 - Northeast: The coefficient for the northeast region is 0.08612. This represents that the increase of 0.08612 in medical charges is associated with living in the northeast, compared to living in the southwest. The coefficient is not significant as indicated by the p-value (0.05206). This suggests that living in the northwest region may not have a meaningful impact on individual medical costs billed by health insurance in the model.
 - Northwest: The coefficient for the northwest region is 0.04365. This represents that the increase of 0.04365 in medical charges is associated with living in the northwest, compared to living in the southwest. The coefficient is not significant as indicated by the p-value (0.32344). This suggests that living in the northwest region may not have a meaningful impact on individual medical costs billed by health insurance in the model.
 - Southeast: The coefficient for the southeast region is 0.00510. This represents that the increase of 0.00510 in individual medical costs billed by health insurance is associated with living in the southeast, compared to living in the southwest. The coefficient is not significant as indicated by the p-value (0.90509). This suggests that living in the southeast region may not have a meaningful impact on individual medical costs billed by health insurance in the model.
 - Southwest: Set as a baseline level to fit the model on the remaining level.

6. Model Validation

Besides the Adjusted R-squared and cross-validation RMSE we discussed previously, we also validate our model by checking the assumption for the linear model.

- Linearity: Figure 17 shows the residual versus fitted values we use to evaluate the linearity assumption. The relationship between each predictor and the response variable should be linear. If the relationship is linear, there should be no systematic pattern in the residuals; they should be randomly distributed around zero lines across all values of the predictors. In Figure 17, we can observe patterns that suggest possible nonlinearity in the relationship between the predictors and the response variable as the residuals display clustering at both ends of the fitted value. This indicates that the current linear model might not adequately capture all the underlying relationships.

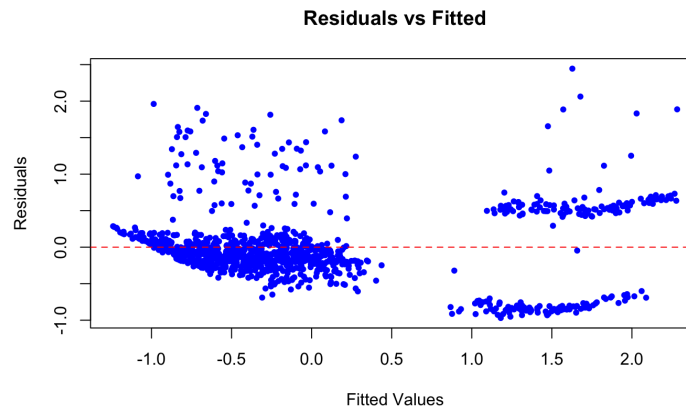


Figure 17. Residual against predicted value

- Independence of Residual: For the validity of the tests used to estimate the coefficient, the residual should be independent of each other. Figure 18 shows the results of the Durbin-Watson test, which tests for autocorrelation in the residual. The test statistic ranges from 0 to 4, where a value closer to 2 suggests no autocorrelation. The Durbin-Watson statistic we obtained is 2.057, which is close to 2. This suggests that there is no autocorrelation in the residual of our model. The null hypothesis is that the residuals are uncorrelated. With a large p-value of 0.5265, we fail to reject the null hypothesis, further confirming that the residuals in our model are uncorrelated.

Durbin-Watson test

```
data: lm_model
DW = 2.057, p-value = 0.5265
alternative hypothesis: true autocorrelation is greater than 0
```

Figure 18. Result of Durbin-Watson Test

- Normality: Figure 19 shows the Q-Q plot to check for normality assumption. If the assumption is held for the model, the points will fall on the 45-degree reference line. In Figure 19, we can observe that the residual generally follows the line in the center portion of the plot. However, there are obvious deviations from the line at both tails. This suggests that while the residuals are approximately normal for the central range of the data, the presence of heavy tails violates our normality assumption.

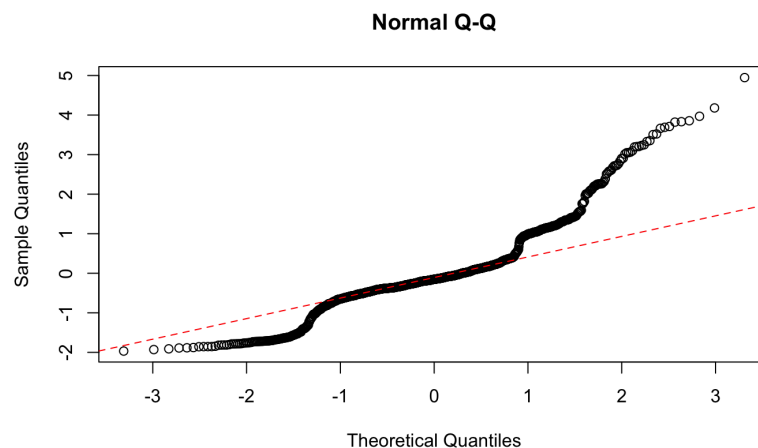


Figure 19. Q-Q Plot

- Homoscedasticity: Figure 20 is the square root of residuals versus fitted values plot. It allows us to check if the variance of the residual changes as fitted values. In Figure 4, we can observe a pattern where the variance of

residuals increases with higher fitted values. This indicates heteroscedasticity as the variance of residuals is not constant across the range of fitted values.

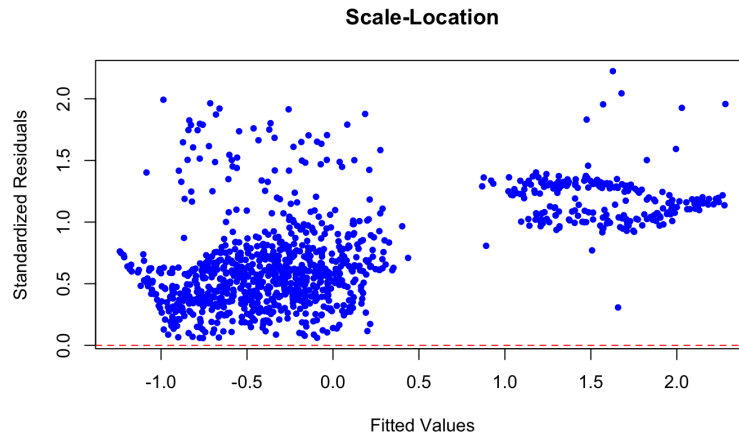


Figure 20. Square Root of Residual vs Fitted Value

Conclusion for validity check:

Given these findings, it may be necessary to consider adjusting the current model or using an alternative modelling technique. We might be able to address such issues by application of different regression techniques that could better manage the challenges observed, such as non-linearity and heteroscedasticity. This will help us to ensure that the final model not only fits the data well but is also capable of providing reliable predictions and insights for decision-making.

7. Hypothesis Test

Summary of linear regression:

```
##
## Call:
## lm(formula = charges ~ smoker, data = insurance)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19221  -5042   -919    3705   31720
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8434.3      229.0   36.83  <2e-16 ***
## smokers       23616.0      506.1   46.66  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7470 on 1336 degrees of freedom
## Multiple R-squared:  0.6198, Adjusted R-squared:  0.6195
## F-statistic: 2178 on 1 and 1336 DF, p-value: < 2.2e-16
```

Figure 21. Summary table for linear regression

This summary table showed that the estimated intercept is 8434.3, which means that when the value of smoker is zero, the expected value of charges for non-smokers is 8434.3. The coefficient of the predictor smokers is 23626. This represents that smokers are charged 23626 more than non-smokers on average. Noticeably, both p-values of intercept and smokers are less than $2e-16$. Therefore, they are statistically significant, and smoking status significantly influences charges. The adjusted R-squared value of 0.6195 implies that 61.95% of the variability in charges is explained by the variable smoker. This suggests that smoking status is a strong contributing factor to charges. Since the p-value of F-statistics is smaller than

2.2e-16, the model is statistically significant. In order to test the overall significance of the predictor smoker and examine variance, we still need an ANOVA table.

Ridge and Lasso linear models all indicate that the predictor smoker has the most significant impact on medical expenses. Hence, we obtained an ANOVA table of the linear regression to conduct a hypothesis test to check whether this assumption holds.

```
## Analysis of Variance Table
##
## Response: charges
##          Df      Sum Sq   Mean Sq F value    Pr(>F)
## smoker      1 1.2152e+11 1.2152e+11  2177.6 < 2.2e-16 ***
## Residuals 1336 7.4554e+10 5.5804e+07
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 22. Anova table

The null hypothesis $H_0: \beta_{\text{smoker}} = 0$, and the alternative hypothesis $H_1: \beta_{\text{smoker}} \neq 0$. From this ANOVA table, the P-value of F-statistics is smaller than 2.2e-16 which is close to zero, suggesting that we reject the null. Therefore, the linear regression model of smoker is statistically significant, and the predictor smoker significantly explained variance in charges. To further compare the group of smokers and non-smokers, we performed a Welch two-sample t-test to test if the means of the two groups were equal.

```
##
## Welch Two Sample t-test
##
## data:  charges by smoker
## t = -32.752, df = 311.85, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -25034.71 -22197.21
## sample estimates:
## mean in group 0 mean in group 1
##      8434.268      32050.232
```

Figure 23. Result of Welch two sample t-test

The null hypothesis of the Welch two-sample t-test is that the means of the two groups are the same. The alternative hypothesis is that those two populations have different means. According to the result of the t-test, the p-value is also less than 2.2e-16, which is extremely small. Therefore, we reject the null hypothesis, meaning that there is a difference between the charges of the two groups. The 95% confidence interval for the mean difference of the two populations is from -25034.71 to -22197.21. This means that we are 95% confident that the difference in means will lie in this range. The negative sign showed that the expense of smokers is much higher than the costs of non-smokers. The average charge of people who smoke is 32050.232, and, in contrast, the mean cost for non-smokers is 8434.268. Therefore, we confirmed that the predictor smoker significantly impacts the medical expenses.

Limitation and Further Research

All of those three models fail to capture complicated relationships between variables because they only identify linear relationships. To address this limitation, we will consider utilizing more advanced techniques such as neural networks to identify non-linear relationships in the future. As shown in Figure 16, the RMSE of the three models is close, which indicates that the three models have similar performance. This typically occurs when the dataset is well-formatted, and its predictors are independent and relevant. Under such circumstances, the benefits of regulations added to ridge and lasso

regression are reduced. Since outliers of predictors are not eliminated, they impact the means of parameters and lead to deviations in regression lines. Elimination of outliers may improve the performance of models.

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